Department: CSE Theme: Dynamic Bandwidth Optimization in CCTV Networks

Adaptive CCTV Compression System Using Dynamic Region of Interest (ROI) Control

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The objective of this research is to develop an adaptive CCTV compression framework capable of intelligently managing video quality, bitrate, and bandwidth by dynamically tuning encoding parameters according to scene activity and regions of interest Region of Interest (ROI). Traditional CCTV systems transmit video at constant bitrates, consuming excessive storage and network resources even in low-motion environments. To address this inefficiency, the proposed system integrates OpenCV-based motion detection with Ffmpeg-driven adaptive encoding, enabling selective bitrate control that prioritizes critical visual zones while suppressing background redundancy .The architecture comprises three synchronized modules: an Edge Node that captures live video streams, performs ROI detection, and applies dynamic compression; a Central Server that manages client connections, logs compression metrics, and exposes a RESTful /metrics endpoint; and a Real-Time Evaluation Dashboard that visualizes bitrate fluctuations, latency, and frame rate trends through automated data polling. The Python-based evaluator (evaluate.py) continuously collects and analyzes performance metrics, performing controlled experiments (e.g., BG10_ROI70, BG50_ROI90) with time-based sampling and CSV-based logging for reproducibility. Empirical results confirm a 45–65% reduction in average bandwidth utilization without significant visual degradation in ROI regions, maintaining consistent Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure(SSIM) scores across tests. The system demonstrates real-time adaptability and resilience, automatically adjusting to variable motion levels and client loads. The framework can be extended with machine learning models for intelligent ROI detection and adaptive bitrate prediction, enabling data driven optimization of video quality and bandwidth utilization .This project establishes a scalable, energy-efficient model for intelligent video streaming in smart surveillance, IoT-enabled monitoring, and edge-based analytics systems, paving the way for future AI-driven compression optimization in bandwidth-constrained environments.